Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Boolean conditions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Boolean conditions trace and record the first step you can explain without help.

Use W2: Trace separate checks for **Boolean** conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Boolean conditions explanation against the supplied facts before

> discussing it.

Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Combine requirements without replacing AND

> Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Translate the requirement

> Translate the requirement. Apply the supplied Boolean conditions facts and keep your reasoning.

Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace separate checks

> Trace separate checks. Apply the supplied Boolean conditions facts and keep your reasoning.

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Distinguish alternatives

> Distinguish alternatives. Apply the supplied Boolean conditions facts and keep your reasoning.

Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Write a new rule independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Write a new rule independently. Apply the supplied Boolean conditions facts and keep your reasoning.

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

Choose a case that detects replacing and with or between the range check and available.

> **My reasoning and evidence**

W5 Separate a check from a guarantee

> Separate a check from a guarantee. Apply the supplied Boolean conditions facts and keep your reasoning.

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Boolean conditions to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Boolean conditions

> Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive. Explain the labelled step before checking the changed condition.

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

AI ACADEMY / PYTHON-BRIDGE/4

Boolean conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Check lower boundary

15>=12 is True.

2. Check upper boundary

15<=15 is also True.

3. Combine with and

Both comparisons are True, so allowed is True.

CHANGE ONE THING

Change age to 16.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both **comparison**s are True, so allowed is True.

> 1. Check lower boundary

2. Check upper boundary

3. Combine with and

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Combine with and Both **comparison**s are True, so allowed is True. A **Boolean** condition represents a yes-or-no result.

And requires both conditions; checking only one boundary would accept values outside the intended interval.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change age to 16. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The result becomes False.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The result becomes False. The lower condition remains True, but the upper condition fails.

Repair a missing building block

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Boolean conditions picture from its labels, then record a relevant limit.

01. Boolean conditions: the case



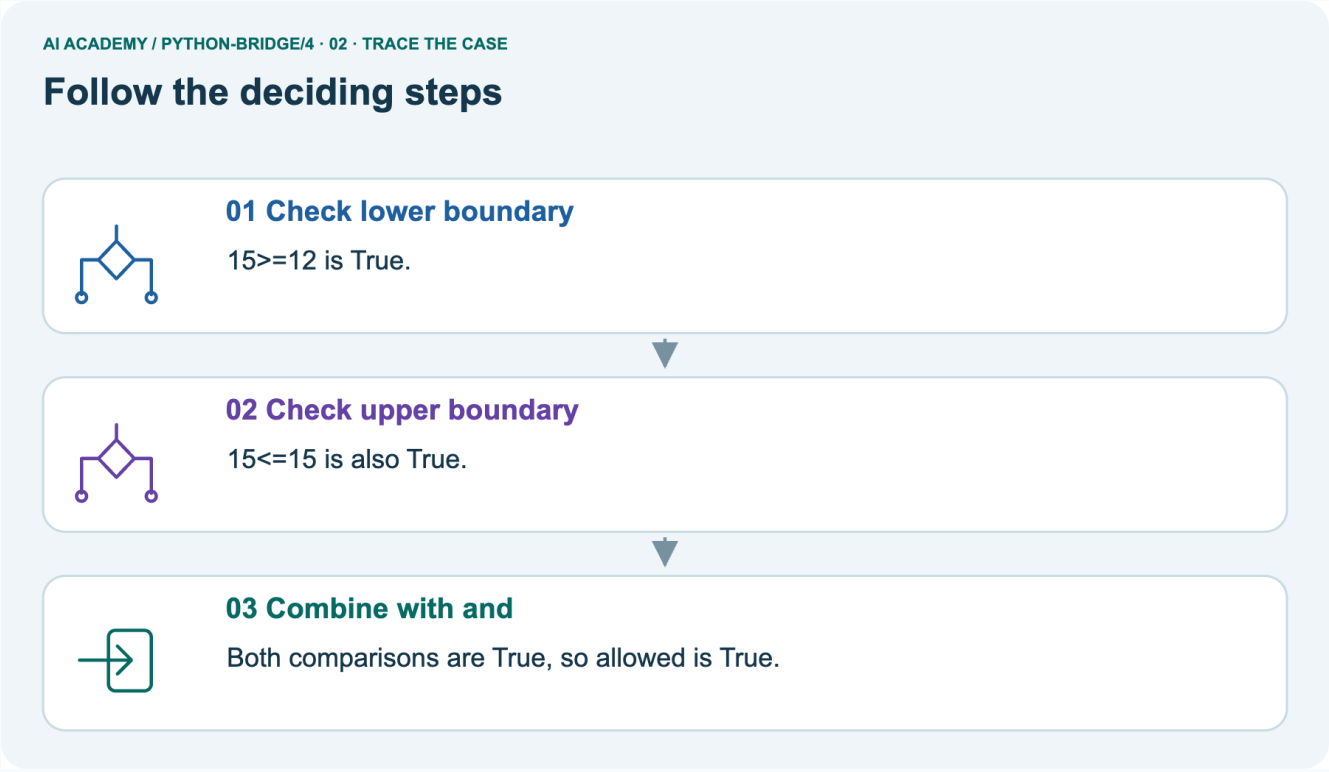
Follow the labelled picture with the matching explanation. Open the vector diagram

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive. Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



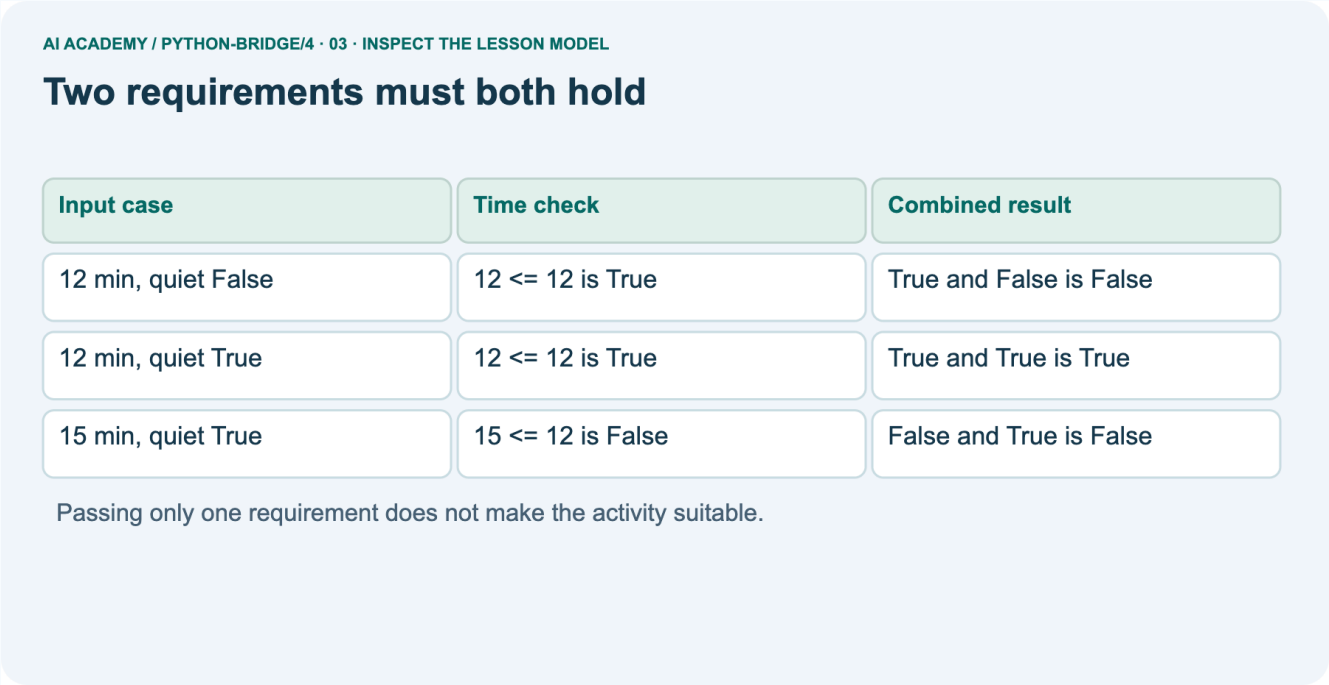
Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two requirements must both hold



Follow the labelled picture with the matching explanation. Open the vector diagram

Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / PYTHON-BRIDGE/4 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Both comparisons are True, so allowed is True.

Changed condition

Change age to 16.

Worked changed-case result

The result becomes False.
Why: The lower condition remains True, but the upper condition fails.

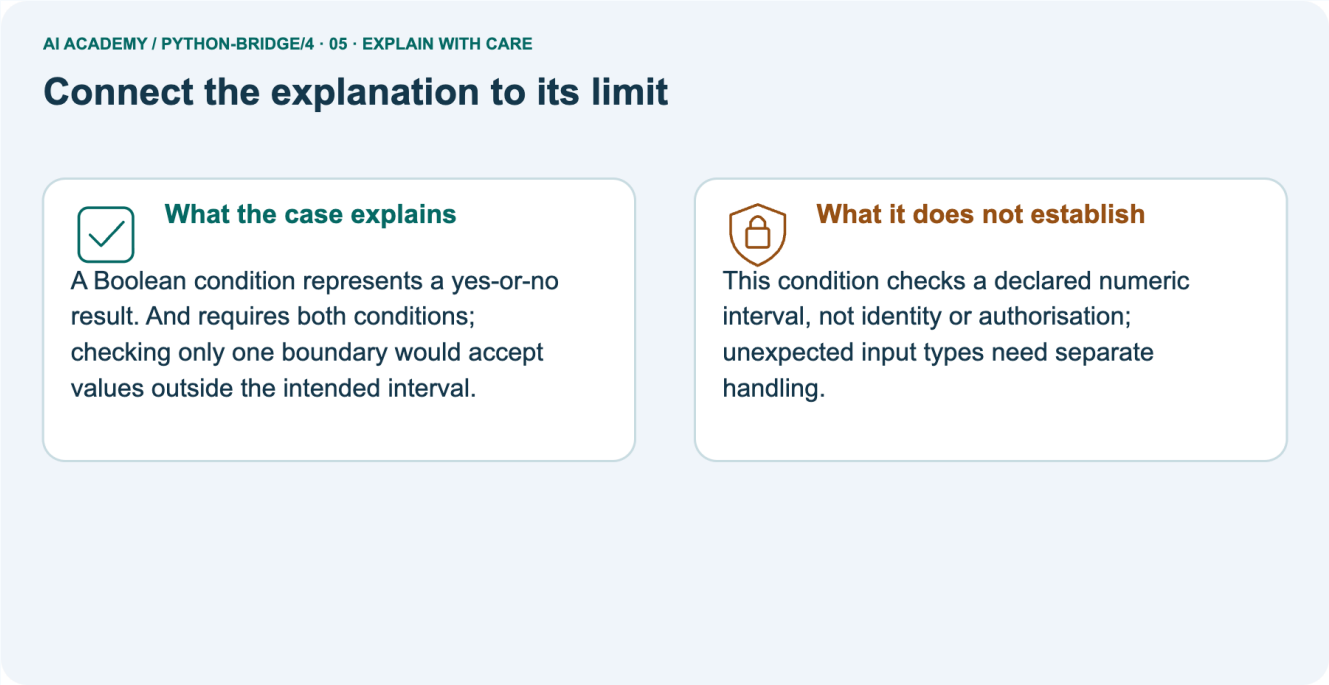
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

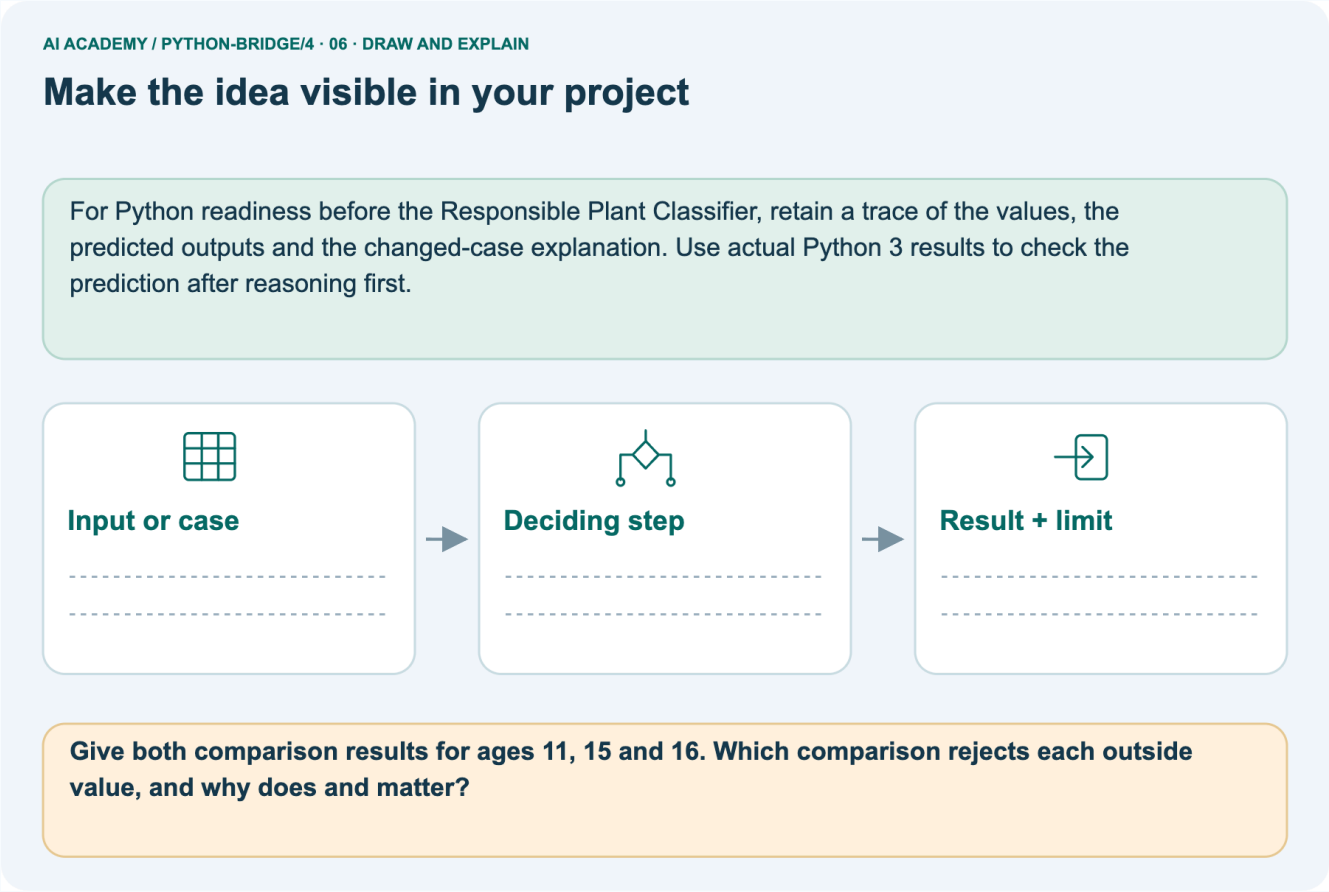
A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.